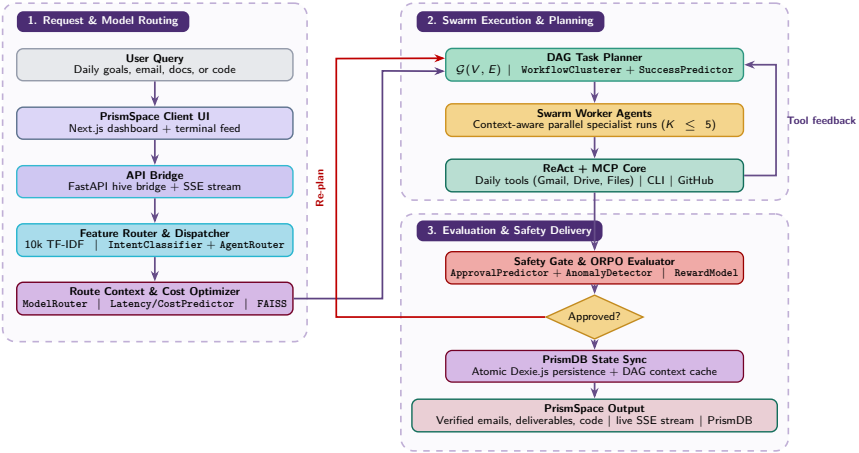
09. PrismSpace System Architecture & Multi-Model Swarm Workflow (Figure 1.2)



11 Models Mapped from `Models/`: **Stage 1:** `intent_classifier` (12-class), `agent_router` (specialist team), `model_router` (OpenAI/Groq/NIM), `latency_predictor`, `cost_predictor`, `faiss.index` (semantic ranker) | **Stage 2:** `workflow_templates` (clusterer), `workflow_success_predictor` (DAG completion) | **Stage 3:** `approval_predictor` (risk score), `anomaly_detector` (IsolationForest), `reward_orpo_v2` (TRL Odds Ratio Alignment)

10. Unified Dataset Preparation & Multi-Model Training Pipeline

1. Multi-Source Raw Datasets

Heterogeneous Ingestion (24 Training Sets):

- **Agent Traces & Telemetry:** *agent-llm-traces* (Exgentic: timings, tool calls), *AgentInstruct* (THUDM: sub-task paths), *cx-cmu–agent_trajectories* (SWE/WebArena), *Open-M3-Bench* (MCP tool states), *tau-bench* (multi-turn retail/airline).
- **Tool & Function Calling:** *APIBench* (Gorilla: HF/TorchHub APIs), *BFCL* (AST/parallel calls), *xlam-function-calling-60k* (60k tool dialogues), *code_generation_lite* (LiveCodeBench), *EnvFactory* (SFT-filtered & RL traces).
- **Retrieval & Domain Knowledge:** *beir* (search corpus), *awesome-harness-engineering* (FAISS schemas), *scientific-agent-skills* (CLI/Bio schemas), *oasst1* (human dialogue trees).
- **Routing, Safety & Preference:** *lmsyschatbot_arena*, *xRouteBench*, *RouteLLM*, *routerbench*, *wildguardmix* (harm labels), *ultrafeedback_binarized* (chosen/rejected), *Qwen2.5-1.5B* (ORPO base weights).

2. Ingestion & Curated Extraction

Step 1: Multi-Format Combiner (`dataset_loader.py`)

- Ingests `.parquet`, `.json`, `.jsonl`, `.pkl`, `.csv`, `.md`.
- Normalizes prompt strings, timestamps, action boundaries, and categorical label spaces into unified schema.

Step 2: Supervised Prep (`prepare_supervised_datasets.py`)

- Generates curated splits in `model/datasets/curated/`:
- `approval/train.jsonl`: WildGuardMix prompt harm labels → binary risk gate.
- `success/train.jsonl`: cx-cmu trajectory end-rewards → task completion predictor.
- `provider/train.jsonl`: Chatbot Arena / RouterBench prompts → OpenAI, Anthropic, Google affinity.
- `reward/train.jsonl`: UltraFeedback pairwise prompt, chosen, rejected completions for alignment.

Step 3: Training Orchestrator (`train.py`)

- Fits 10 scikit-learn, FAISS & TF-IDF models → `.joblib`, `.index`, `.pkl`.
- Fine-tunes `RewardModel` via TRL ORPO on Qwen2.5.

3. Artifacts & Holdout Evaluation

Production Artifacts in `Models/` (11 Models):

- `intent_classifier.joblib`: 12 task intents (Acc 99.6%).
- `agent_router.joblib`: Multi-label team selection.
- `workflow_templates.pkl`: K-Means DAG clustering.
- `workflow_success_predictor.joblib`: DAG completion.
- `model_router.joblib`: Groq / OpenAI / NIM affinity.
- `latency_predictor` & `cost_predictor`: Run telemetry.
- `anomaly_detector.joblib`: IsolationForest baseline.
- `approval_predictor.joblib`: Action safety risk gate.
- `faiss.index` + `vectorizer`: Fast semantic context.
- `reward_orpo_v2`: Qwen2.5-1.5B ORPO preference adapter.

Multi-Tier Rigorous Evaluation:

- `evaluate_models.py`: Evaluates 20% held-out curated test sets (`test.jsonl`) for F1, Precision, Recall, Calibration.
- `evaluate_holdout.py` **(Zero-Leakage Generalization):**
- **GAIA Benchmark (Val + Test):** Multimodal tool-agent reasoning across real complex tasks.
- **SWE-bench Verified (Test):** Real-world GitHub software patches & coding agent routing verification.

End-to-End Pipeline Coherence: Raw multi-modal traces & benchmarks $\xrightarrow{\texttt{dataset_loader}}$ Normalized schemas $\xrightarrow{\texttt{prepare_supervised}}$ Target `.jsonl` splits $\xrightarrow{\texttt{train.py}}$ 11 Production Models $\xrightarrow{\texttt{evaluate}}$ Held-out GAIA & SWE-bench Verified